

Methodology of Econometrics

There are several school of thoughts on econometrics methodology, we present here the "traditional" or "classical" econometrics methodology.

Broadly speaking, traditional econometric methodology proceeds along the following lines:

1. Statement of theory or hypothesis:

The fundamental psychological law.. is that men [women] are disposed, as a rule and on average, to increase their consumption as their income increases but not as much increases in their income.

$C \uparrow$ $Y \uparrow$ having positive relationship.

2. Specification of mathematical model of consumption:-

Keynesian consumption function:

$$0 < \beta_2 < 1$$

$$Y = \beta_1 + \beta_2 X$$

where $Y \rightarrow$ consumption (Independent)

$X =$ income and β_1 and β_2 are "parameters"

Intercept (β_1) and slope (β_2) coefficient.

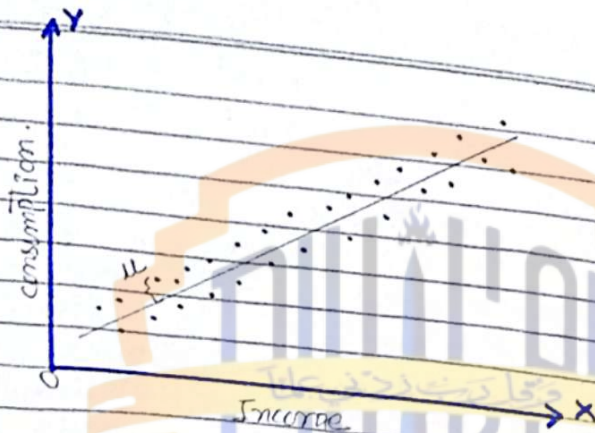
3. Specification of Economic model of consumption:

function as:

$$Y = \beta_1 + \beta_2 X + u$$

consumption

where " u " known as disturbance, error term, stochastic variables (other factors).



4- Obtaining data:

To obtain the numerical values of B_1 and B_2 , we need data.

There are many types of data:

- 1- Time series data (weekly, monthly)
- 2- Cross sectional data (In section form)
- 3- Panel data / Pooled data.

⇒ (combination of Time series or cross sectional data)

- 4- Qualitative data / Dumi variables:

To measures Qualitative data we use dumi variables. "(categorical variables)"

- 5- Engineering data:-

- represent factor of production
- associate with production function.

5. Estimation:

Now estimate the parameters of consumption function. The statistical techniques of

"regression analysis" is the main tool used to obtain the following estimates B_1 and B_2 .

$\hat{Y}_i = -299.5913 + 0.7218X_i$
The hat on the Y estimates that it is on estimates.

6. Hypothesis testing:

For testing our data we uses:

- T-test
- Z-test
- F-Test
- Anova
- Chi-squares.

7. Forecasting: (Prediction)

In which we apply values on model and find "C"

8. Policies:

The all results are written in policy.

"Regression Analysis"

"The method of Ordinary Least Squares" (OLS)

PRF: $Y_i = \beta_0 + \beta_1 X_i + u_i$

SRL: $Y_i = \hat{\beta}_0 + \hat{\beta}_1 X_i + \hat{u}_i$

now $Y_i = \hat{Y} + \hat{u}_i \Rightarrow \hat{u}_i = Y_i - \hat{Y}_i$
where $Y_i \rightarrow$ total value and
 $\hat{Y}_i \rightarrow$ estimated values.

using: $\hat{u}_i = Y_i - \hat{Y}$

⇒ apply " Σ " on both sides.
 $\Sigma \hat{u}_i = \Sigma (Y_i - \hat{Y}_i)$

⇒ which states that SRF can be fixed in such a way that:

$$\sum u_i^2 = \sum (y_i - \hat{y}_i)^2$$

where $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_i$

$$\sum u_i^2 = \sum (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$

⇒ Taking derivative:

Taking derivative w.r.t β_0

$$\frac{d \sum u_i^2}{d \beta_0} = \frac{d}{d \beta_0} \left(\sum (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2 \right)$$

$$\Rightarrow 2 \sum (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) (-1) = 0$$

$$\Rightarrow -2 \sum (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0$$

Now as 2 multiply on right side is zero and equation is:

$$\sum (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0$$

⇒ as $\beta_0 \rightarrow$ constant so write "n" with β_0 as:

$$\sum y_i - n\hat{\beta}_0 - \hat{\beta}_1 \sum x_i = 0$$

$$\sum y_i = n\hat{\beta}_0 + \hat{\beta}_1 \sum x_i$$

$$\boxed{\sum y_i = n\hat{\beta}_0 + \hat{\beta}_1 \sum x_i} \rightarrow (1)$$

⇒ Again derivative on b/s:

again derivative w.r.t " β_1 "

$$\frac{d \sum u_i^2}{d \beta_1} = \frac{d}{d \beta_1} \left(\sum (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2 \right)$$

$$\Rightarrow 2 \sum (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) (-x_i) = 0$$

$$-2 \sum (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) (x_i) = 0$$

2 multiply by 0

$$\Rightarrow \sum (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) (x_i) = 0$$

$$\sum (x_i y_i - \hat{\beta}_0 x_i - \hat{\beta}_1 x_i^2) = 0$$

$$\sum x_i y_i - \hat{\beta}_0 \sum x_i - \hat{\beta}_1 \sum x_i^2 = 0$$

$$\sum x_i y_i = \hat{\beta}_0 \sum x_i + \hat{\beta}_1 \sum x_i^2$$

now using equation (1)

$$\sum y_i = n\hat{\beta}_0 + \hat{\beta}_1 \sum x_i$$

⇒ Dividing b/s by "n"

$$\frac{\sum y_i}{n} = \frac{n\hat{\beta}_0}{n} + \frac{\hat{\beta}_1 \sum x_i}{n}$$

$$\bar{y} = \hat{\beta}_0 + \hat{\beta}_1 \bar{x}$$

$$\boxed{\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}}$$

To Find β_1 :

Using:

i - $\sum y_i = n\hat{\beta}_0 + \hat{\beta}_1 \sum x_i$

ii - $\sum x_i y_i = \hat{\beta}_0 \sum x_i + \hat{\beta}_1 \sum x_i^2$

Equation "1" multiply by " $\sum x_i$ "

$$\sum x_i (\sum y_i = n\hat{\beta}_0 + \hat{\beta}_1 \sum x_i)$$

$$\sum x_i \sum y_i = n\hat{\beta}_0 \sum x_i + \hat{\beta}_1 (\sum x_i)^2$$

⇒ and equation "2" multiply by "n"

$$n(\sum x_i y_i = \beta_0 \sum x_i + \beta_1 \sum x_i^2)$$

$$n \sum x_i y_i = n \beta_0 \sum x_i + n \beta_1 \sum x_i^2$$

⇒ now subtracting both equation

$$\sum x_i \sum y_i = n \beta_0 \sum x_i + \beta_1 (\sum x_i)^2$$

$$n \sum x_i y_i = n \beta_0 \sum x_i + n \beta_1 \sum x_i^2$$

$$\sum x_i \sum y_i - n \sum x_i y_i = \beta_1 (\sum x_i)^2 - n \beta_1 \sum x_i^2$$

(B₁ common)

$$\sum x \sum y - n \sum xy = \beta_1 ((\sum x)^2 - n \sum x^2)$$

now — common on b/s.

$$\sum x_i \sum y_i + n \sum x_i y_i = \beta_1 (-(\sum x)^2 + n \sum x^2)$$

$$\sum x_i \sum y_i + n \sum x_i y_i = \beta_1 ((\sum x)^2 + n \sum x^2)$$

$$\boxed{\beta_1 = \frac{n \sum x_i y_i - \sum x_i \sum y_i}{n \sum x^2 - (\sum x)^2}}$$

Question # 01:-

x	y	xy	x ²
Fertilizers	Yields		
100	40	400	10,000
200	50	10,000	40,000
300	50	15,000	90,000
400	70	28,000	16,000
500	65	32,500	250,000
600	65	39,000	360,000
700	80	56,000	49,000
$\sum x = 2800$	$\sum y = 420$	$\sum xy = 184500$	$\sum x^2 = 140,000$

formula:- $\beta_0 = \bar{y} - \beta_1 \bar{x}$

$$\beta_1 = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2}$$

now putting values as:

$$\beta_1 = \frac{7(184500) - (2800)(420)}{7(140000) - (2800)^2}$$

$$= \frac{1291500 - 1176000}{980000 - 784000} \Rightarrow \frac{115500}{196000}$$

$$\beta_1 = 0.0589$$

$$\bar{y} = \frac{\sum y}{n} \Rightarrow \frac{420}{7} \Rightarrow 60$$

$$\bar{x} = \frac{\sum x}{n} \Rightarrow \frac{2800}{7} \Rightarrow 400$$

Put these values in formula to find β_0 .

$$B_0 = \bar{Y} - \beta_1 \bar{X}$$

$$\beta_0 = 60 - (0.0589)(400)$$

$$= 60 - 23.56$$

$$= 60 - 23.56$$

$$\beta_0 = 60 - 23.56 \Rightarrow 60 - 23.56 \Rightarrow 36.44$$

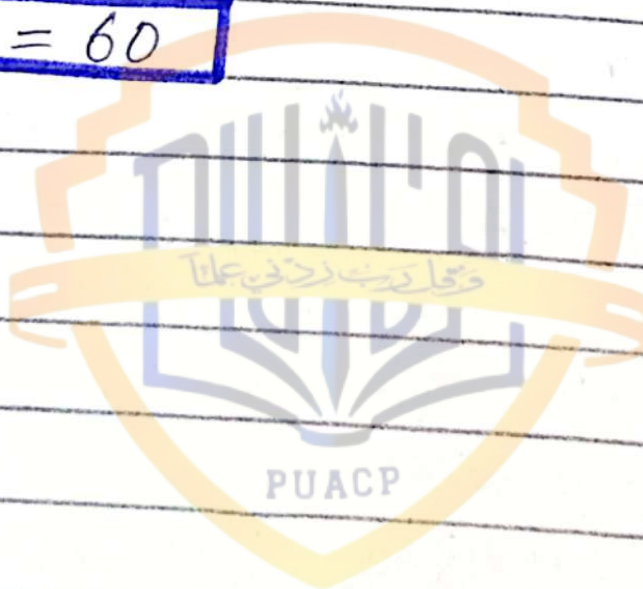
now

$$\hat{Y} = 36.44 + 0.0589X$$

So as $X = 400$ then:

$$\hat{Y} = 36.44 + 0.0589(400)$$

$$\hat{Y} = 60$$



Question # 02:-

X	Y	XY	X ²	ΣX = 424
15	60	900	225	
24	45	1080	576	ΣY = 363
25	50	1250	625	
30	35	1050	900	ΣXY = 12815
35	42	1470	1225	
40	46	1840	1600	ΣX ² = 21926
45	28	1260	2025	
65	20	1300	4225	n = 10
70	22	1540	4900	
75	15	1125	5625	ȳ = ΣY / n = 363 / 10 = 36.3

Formulas:-

$$\beta_0 \therefore \bar{Y} - \beta_1 \bar{X}$$

$$\beta_1 \therefore \frac{n \sum XY - \sum X \sum Y}{n \sum X^2 - (\sum X)^2}$$

Substituting values:-

$$= \frac{10(12815) - (424)(363)}{10(21926) - 179776}$$

$$= \frac{128150 - 153912}{219260 - 179776} \Rightarrow \frac{25762}{39484}$$

$$\beta_1 = -0.6524$$

$$\beta_0 = \bar{Y} - \beta_1 \bar{X}$$

$$= 36.3 - (-0.6524)(42.4)$$

$$= 36.3 + 27.6645$$

$$\beta_0 = 63.96$$

$$\hat{Y} = \beta_0 - \beta_1 X$$

$$= 63.96 - (-0.6524)X \Rightarrow 63.96 + 0.6524X$$

$$\hat{Y} = 63.96 + 0.6524X$$

$$\text{Put } X = 9750$$

$$\hat{Y} = 63.96 + 0.6524(9750)$$

$$\hat{Y} = 6424.86$$

Question # 03:-

weight	chut.s	XY	X ²
2.71	29.5	79.945	7.3441
2.15	26.3	56.545	4.6225
4.40	32.2	141.68	19.36
8.50	36.5	200.75	30.25
3.20	27.2	87.04	10.24
4.30	27.5	118.25	18.49
2.31	28.3	65.373	5.3361
4.30	30.3	130.29	18.49
3.70	28.5	105.45	13.69
$\Sigma X = 32.57$	$\Sigma Y = 266.3$	$\Sigma XY = 985.323$	$\Sigma X^2 = 127.8227$

Formula:-

$$B_0 = \bar{Y} - \beta_1 \bar{X}$$

$$\beta_1 = \frac{n \Sigma XY - \Sigma X \Sigma Y}{n \Sigma X^2 - (\Sigma X)^2}$$

$$= \frac{(9)(985.323) - (32.57)(266.3)}{9(127.8227) - (32.57)^2}$$

$$= \frac{8867.907 - 8673.391}{150.4043 - 1060.8049} \Rightarrow \frac{194.516}{-289.8994}$$

$$\beta_1 = 0.6716$$

$$\bar{Y} = \frac{\Sigma Y}{n} = \frac{266.3}{9} = 29.59$$

$$\bar{X} = \frac{\Sigma X}{n} = \frac{32.57}{9} = 3.619$$

$$B_0 = \bar{Y} - \beta_1 \bar{X}$$

$$= 29.59 - (0.6716)(3.619)$$

$$B_0 = 29.59 - 2.4307$$

$$B_0 = 26.28$$

$$\hat{Y} = B_0 - \beta_1 X \Rightarrow \hat{Y} = 26.28 - 0.9140X$$

$$\hat{Y} = 26.28 - 0.9140X$$

$$\text{Put } X = 500$$

$$\hat{Y} = 26.28 - 0.9140(500)$$

$$\hat{Y} = 21.71$$

Question #4:-

^x Paper I	^y Paper II	xy	x^2
45	56	2520	2025
55	50	2750	3025
56	48	2688	3136
58	60	3480	3364
60	62	3720	3600
65	64	4160	4225
68	65	4420	4624
70	70	4900	4900
75	74	5550	5625
80	82	6560	6400
85	90	7650	7225
$\Sigma x = 717$	$\Sigma y = 721$	$\Sigma xy = 48.398$	$\Sigma x^2 = 48.149$

Formula:- $\beta_0 = \bar{y} - \beta_1 \bar{x}$

$$\begin{aligned} \beta_1 &= \frac{n \Sigma xy - \Sigma x \Sigma y}{n \Sigma x^2 - (\Sigma x)^2} \Rightarrow \frac{11(48.398) - (717)(721)}{529639 - 514089} \\ &= \frac{532378 - 516957}{529639 - 514089} \Rightarrow \frac{15421}{15550} = 0.9917 \end{aligned}$$

$$\beta_0 = \bar{y} - \beta_1 \bar{x}$$

$$\bar{y} = \frac{\Sigma y}{n} = \frac{721}{11} \Rightarrow 65.545$$

$$\bar{x} = \frac{\Sigma x}{n} = \frac{717}{11} \Rightarrow 65.18$$

$$\begin{aligned} \beta_0 &= \bar{y} - \beta_1 \bar{x} \\ &= 65.545 - (0.9917)(65.18) \end{aligned}$$

$$\beta_0 = 65.545 - 64.641$$

$$\beta_0 = 0.904$$

$$\hat{Y} = \beta_0 - \beta_1 X$$

$$\hat{Y} = 0.904 - 0.9917 X.$$

$$\text{Put } X = 0.5$$

$$\hat{Y} = 0.904 - 0.9917 (0.5)$$

$$= 0.904 - 0.495 \Rightarrow 0.409$$



"Deviation form"

Property 1:- (linearity)

$$\hat{\beta} = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2} \quad \text{now dividing by } n$$

$$\frac{n \sum xy / n - \sum x \sum y / n}{n \sum x^2 / n - (\sum x)^2 / n} \Rightarrow \frac{\sum xy - \frac{\sum x \sum y}{n}}{\sum x^2 - \frac{(\sum x)^2}{n}}$$

$$= \frac{\sum xy - \bar{x} \bar{y}}{\sum x^2 - (\bar{x})^2} \Rightarrow \frac{\sum [(x - \bar{x})(y - \bar{y})]}{\sum (x - \bar{x})^2}$$

$$\hat{\beta} = \frac{\sum xy}{\sum x^2} \quad \text{Let } x - \bar{x} = x$$

$$\text{in deviation form } \Rightarrow \hat{\beta} = \frac{\sum xy}{\sum x^2} = \frac{\sum x(y - \bar{y})}{\sum x^2}$$

$$\hat{\beta} = \frac{\sum xy - \sum x \bar{y}}{\sum x^2}$$

$$\hat{\beta} = \frac{\sum xy - \sum \bar{y}(x - \bar{x})}{\sum x^2} \quad \because x = x - \bar{x}$$

$$\hat{\beta} = \frac{\sum xy - 0}{\sum x^2} \Rightarrow \hat{\beta} = \frac{\sum xy}{\sum x^2} \quad \because x - \bar{x} = 0$$

$$\text{Let } \therefore \frac{\sum x}{n^2} = k \quad \text{So,}$$

$$\hat{\beta} = \sum ky \quad \text{Linear function of dependent variable.}$$

Question # 01:-

according to this formula:-

$$\hat{\beta} = \frac{\sum xy}{\sum x^2}$$

X	Y	$x = x - \bar{x}$	$y = y - \bar{y}$	xy	x^2
100	40	-300	-20	6000	90,000
200	50	-200	-10	2000	40,000
300	50	-100	-10	1000	10,000
400	70	0	10	0	0
500	65	100	5	500	10,000
600	65	200	5	1000	40,000
700	80	300	20	6000	90,000
2800	420	0	0	16500	280,000

$$\bar{x} = \frac{\sum x}{n} = 400$$

Formula:-

$$\hat{\beta} = \frac{\sum xy}{\sum x^2} \Rightarrow \frac{16500}{280,000} = 0.0589$$

Question #02:-

X	Y	$x = X - \bar{X}$	$y = Y - \bar{Y}$	xy	x^2
15	60	-27.4	23.7	-649.38	750.76
24	45	-18.4	8.7	-160.08	338.56
25	50	-17.4	13.7	-238.38	302.76
30	35	-12.5	-1.3	16.25	156.25
40	42	-2.4	8.7	-21.18	5.76
45	46	2.6	9.7	25.28	6.76
65	28	22.6	-8.3	-187.58	510.76
70	20	27.6	-16.3	-450.38	761.76
75	22	32.6	14.3	466.18	1062.76
35	15	-7.4	21.3	-157.18	54.76
424	363	-	-	-397.95	3.956

$$\bar{X} = \frac{\sum X}{n} = \frac{424}{10} = 42.4 \quad \bar{Y} = \frac{363}{10} = 36.3$$

$$\hat{\beta} = \frac{\sum xy}{\sum x^2} = \frac{-397.95}{3.956}$$

$$\hat{\beta} = -100.594$$

Question #03:-

X	Y	$x = X - \bar{X}$	$y = Y - \bar{Y}$	xy	x^2
2.75	29.5	-0.873	-0.08	0.06984	0.762
2.15	26.3	-1.473	-3.28	4.83144	2.169
4.40	32.2	+0.77	2.62	2.0174	0.592
5.50	36.5	+1.877	6.92	12.9884	3.523
3.20	27.2	-0.423	-2.38	1.00674	0.178
4.30	27.5	0.677	-2.08	-1.40816	0.458
2.31	28.3	-1.313	-1.28	1.68064	1.723
4.30	30.3	0.677	0.72	0.48744	0.458
3.70	28.5	0.077	-1.08	-0.08316	0.008
32.61	266.3	-	-	21.59	9.868

$$\bar{X} = \frac{\sum X}{n} = \frac{32.61}{9} = 3.623$$

$$\bar{Y} = \frac{\sum Y}{n} = \frac{266.3}{9} = 29.58$$

$$\text{formula-} \quad \frac{\sum xy}{\sum x^2} \Rightarrow \frac{21.59}{9.868} = 2.187$$

$$\hat{\beta} = 2.1878$$

Question # 04:-

X	Y	$x = x - \bar{x}$	$y = y - \bar{y}$	xy	x^2
45	56	-20.18	-9.54	192.51	407.23
55	50	-10.18	-15.54	158.197	103.63
56	48	-9.18	-17.54	161.01	84.27
58	60	-7.18	-5.54	39.77	51.55
60	62	-5.18	-3.54	18.33	26.83
65	64	-0.18	-1.54	0.277	0.032
68	65	2.82	0.54	1.522	7.952
70	70	4.82	4.46	21.49	23.23
75	74	9.82	8.46	83.07	96.43
80	82	14.82	16.46	243.93	219.63
85	90	19.82	24.46	484.79	392.83
$\Sigma x = 717$	$\Sigma y = 721$	-	-	1404.926	1413.614

$$\bar{x} = \frac{\Sigma x}{n} = \frac{717}{11} = 65.18, \quad \bar{y} = \frac{\Sigma y}{n} = \frac{721}{11} = 65.54$$

Formula:-

$$\hat{\beta} = \frac{\Sigma xy}{\Sigma x^2} \rightarrow \frac{1404.926}{1413.614} = 0.993$$

Property # 02:-

⇒ Unbiasedness:

Proof:-

$$E(\hat{\beta}) = \beta$$

$$Y_i = \alpha + \beta X + u \quad \text{--- (1)}$$

Apply Σ on b/s.

$$\Sigma Y_i = \Sigma \alpha + \beta \Sigma X + \Sigma u_i$$

$$\Sigma Y = n\alpha + \beta(\Sigma x) + \Sigma u_i$$

dividing b/s by "n" as.

$$\frac{\Sigma Y}{n} = \frac{n\alpha}{n} + \frac{\beta(\Sigma x)}{n} + \frac{\Sigma u_i}{n} \quad \therefore \Sigma u = 0$$

$$\bar{y} = \alpha + \beta \bar{x} + 0$$

$$Y = \alpha + \beta \bar{x} \quad \text{--- (2)}$$

Substituting equation (2) from equation (1)

$$(Y - \bar{y}) = \alpha - \alpha + \beta(x - \bar{x}) + u$$

$$y = \beta x + u$$

$$\text{as } y - \bar{y} = y, \quad x - \bar{x} = x$$

$$\text{as we know that } \hat{\beta} = \frac{\Sigma xy}{\Sigma x^2}$$

$$y = \beta x + u$$

$$\hat{\beta} = \frac{\Sigma x(\beta x + u)}{\Sigma x^2}$$

$$\hat{\beta} = \frac{\beta \Sigma x^2 + \Sigma xu}{\Sigma x^2} \Rightarrow \frac{\beta \Sigma x^2}{\Sigma x^2} + \frac{\Sigma xu}{\Sigma x^2}$$

$$\therefore \Sigma u = 0$$

$$\hat{\beta} = \frac{\beta \sum x_i^2 + 0}{\sum x_i^2} \Rightarrow \boxed{\hat{\beta} = \beta}$$

Expected $\hat{\beta}$ = population β .
 now apply expectations. or,

$$\boxed{E(\hat{\beta}) = \beta} \quad \text{here proved.}$$

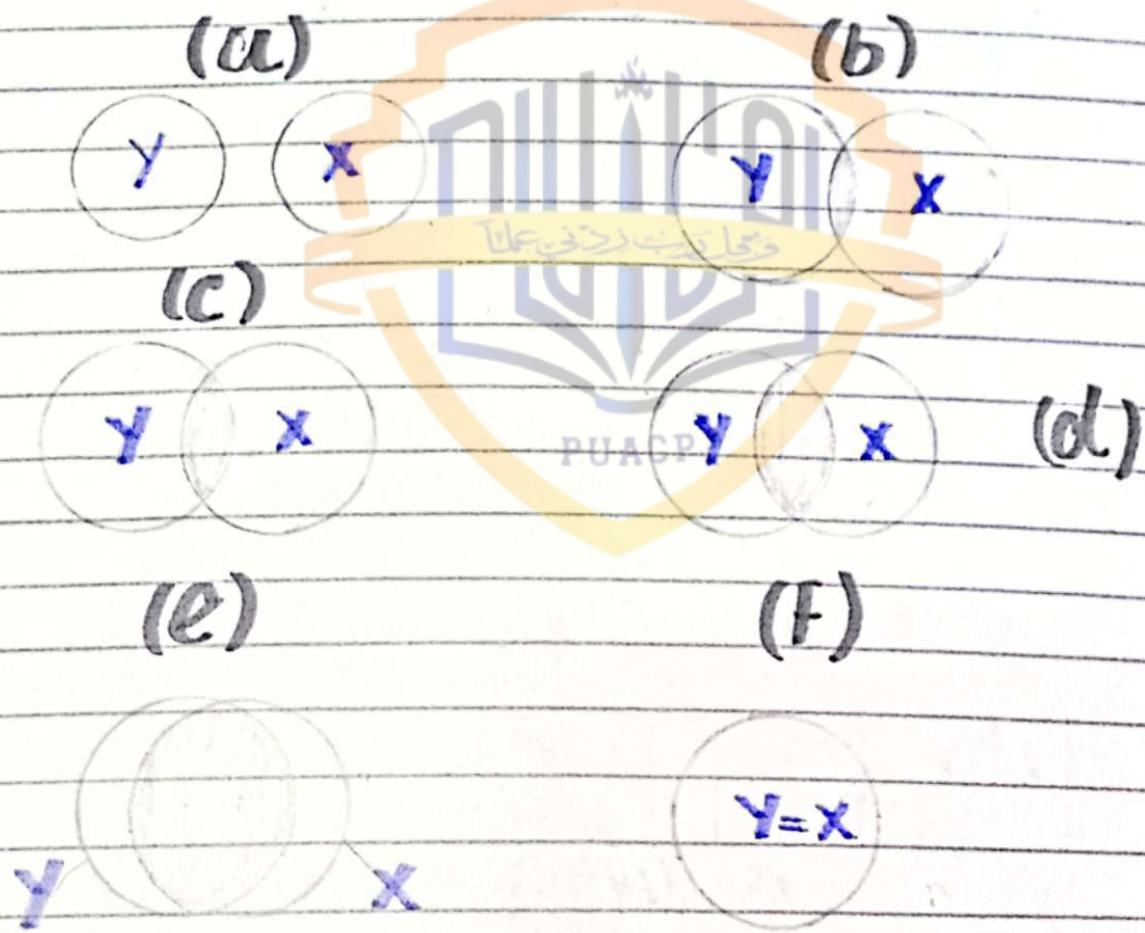
Hence proved that the estimated data
 are unbiased.

"Goodness of fit" Coefficient of Determination

"Total variation / Explained sum of square / Un-explained sum of square"

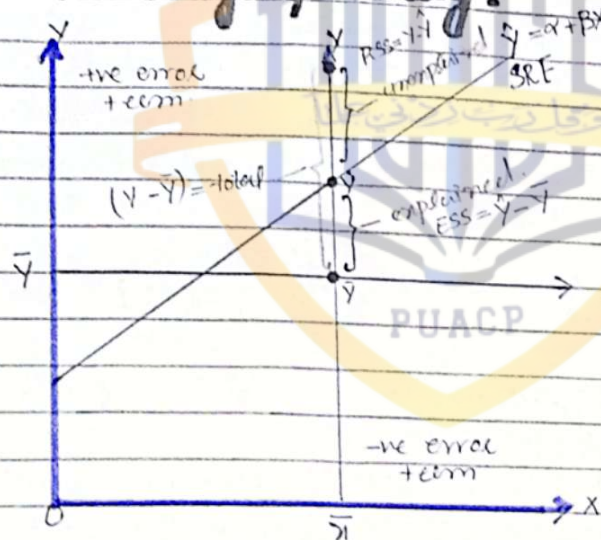
The coefficient of determination is a summary of measures that tells how well the sample regression line fits the data.

Before we show how r^2 is computed, let us consider a explanation of r^2 in terms of a graphical device known as the "ven diagram" as shown is figure as:



In this figure the circle Y represent variation in the "independent variable" and circle X represent variation in the "explanatory variable" X. The "shaded area" indicates the extent to which the variation in Y is explained by the variation in X.

represent graphically:



$$RSS = Y - \hat{Y}, \quad ESS = \hat{Y} - \bar{Y}, \quad TSS = Y - \bar{Y}$$

$$TSS = ESS + RSS$$

$$(Y - \bar{Y})^2 = (\hat{Y} - \bar{Y})^2 + (Y - \hat{Y})^2$$

Thus our formula is as:

$$r^2 = \frac{ESS}{TSS}$$

formula of coefficient sum of square.

and r^2 lies — 0 or 1

$$\text{as } r^2 = \frac{ESS}{TSS}$$

substituting ESS values and TSS values that find

from graph as:

$$r^2 = \frac{ESS}{TSS} = \frac{(\hat{Y} - \bar{Y})^2}{(Y - \bar{Y})^2}$$

$$\text{as } TSS = \sum (Y - \bar{Y})^2 \text{ multiply by } \sum$$

$$TSS = \sum (Y - \bar{Y})^2$$

and also ESS multiply by $\sum$ then.

$$ESS = \sum (\hat{Y} - \bar{Y})^2$$

$$TSS = \sum (Y - \bar{Y})^2, \quad ESS = \sum (\hat{Y} - \bar{Y})^2$$

$$= \sum Y^2 - \sum (\bar{Y})^2, \quad = \sum (\hat{Y})^2 - \sum (\bar{Y})^2$$

$$= \sum Y^2 - (\sum Y/n)^2, \quad \text{as } \hat{Y} = \alpha + \beta X \text{ so,}$$

$$= \sum Y^2 - (\sum Y/n)^2 = \sum (\alpha + \beta X)^2 - (\sum Y/n)^2$$

$$RSS = \sum (Y - \hat{Y})^2$$

$$= \sum (Y)^2 - \sum (\hat{Y})^2 \Rightarrow \sum (Y)^2 - \sum (\alpha + \beta X)^2$$

$$= \sum (Y)^2 - \sum (\alpha + \beta X)^2$$

$$R^2 = \frac{ESS}{TSS} = \frac{\sum (\alpha + \beta X)^2 - (\sum Y/n)^2}{\sum Y^2 - (\sum Y/n)^2}$$

$$\text{as } \alpha + \beta X = \hat{Y} \text{ so,}$$

$$= \frac{\sum (\hat{Y})^2 - (\sum Y/n)^2}{\sum Y^2 - (\sum Y/n)^2}$$

$\Rightarrow$ now multiply $\sum Y$ regression line as:

$$\text{As } \sum \hat{Y} (\hat{Y} = \alpha + \beta X)$$

$$\sum Y \hat{Y} = \alpha \sum Y + \beta \sum XY$$

$$\sum \hat{Y}^2 = \alpha \sum \hat{Y} + \beta \sum X Y$$

Using:

$$\sum (\hat{Y})^2 - (\sum Y/n)^2$$

substituting

$$= \frac{\sum (Y)^2 - (\sum Y/n)^2}{\sum Y^2 - (\sum Y/n)^2} \text{ the value of } \sum (\hat{Y})^2:$$

$$= \frac{\alpha \sum \hat{Y} + \beta \sum X Y - (\sum Y/n)^2}{\sum Y^2 - (\sum Y/n)^2}$$

This is the r^2 formulas as:

$$r^2 = \frac{\sum \hat{\alpha} \sum Y + \hat{\beta} \sum XY - (\sum Y/n)^2}{\sum Y^2 - (\sum Y/n)^2}$$

We also using regression formula as:

$$r^2 = \frac{n \sum xy - (\sum x)(\sum y)}{[n \sum x^2 - (\sum x)^2][n \sum y^2 - (\sum y)^2]}$$

coefficient regression formula.

$$r^2 = \frac{n \sum xy - (\sum x)(\sum y)}{[n \sum x^2 - (\sum x)^2][n \sum y^2 - (\sum y)^2]}$$

Question # 01:-

$$n = 7, \sum xy = 184500, \sum x = 2800, \sum y = 420, \sum x^2 = 140000, \sum y^2 = 26350$$

formulas:-

$$r^2 = \frac{n \sum xy - (\sum x)(\sum y)}{[n \sum x^2 - (\sum x)^2][n \sum y^2 - (\sum y)^2]}$$

$$r^2 = \frac{7(184500) - (2800)(420)}{[7(140000) - (2800)^2][7(26350) - (420)^2]}$$

$$r^2 = \frac{1291500 - 1176000}{[980000 - 784000][124450 - 176400]}$$

$$r^2 = \frac{115500}{(196000)(8050)}$$

$$r^2 = \frac{115500}{0.00000000015}$$

$$\Rightarrow r^2 = 0.00000000007$$

Problem No 1:-

x	y	xy	x ²	y ²
80	70	5600	6400	4900
100	65	6500	10000	4225
120	90	10800	14400	8100
140	95	13300	19600	9025
160	110	17600	25600	12100
180	115	20700	32400	13225
200	120	24000	40000	14400
220	140	30800	48400	19600
240	155	37200	57600	24025
260	150	39000	67600	22500

$$\sum x = 1700, \sum y = 1120, \sum xy = 205500, \sum x^2 = 322000, \sum y^2 = 132100, n = 10$$

Formula of r^2 :-

$$r^2 = \frac{n \sum xy - (\sum x)(\sum y)}{[n \sum x^2 - (\sum x)^2][n \sum y^2 - (\sum y)^2]}$$

$$r^2 = \frac{10(205500) - (1700)(1110)}{[10(322000) - (1700)^2][10(132100) - (1110)^2]}$$

$$r^2 = \frac{2055000 - 1887000}{(3220000 - 2890000)(1321000 - 1232100)}$$

$$\Rightarrow r^2 = \frac{168000}{(336000)(88900)}$$

$$r^2 = 0.000005$$

$$r^2 = 0.000005$$

Problem No 2:-

X	Y	XY	X ²	Y ²
18	77	1386	324	5929
16	78	1248	256	6084
15	79	1185	225	6241
15	79	1185	225	6241
12	80	960	144	6400
10	80	800	100	6400
10	81	810	100	6561
7	82	574	49	6724
9	84	756	81	7056
6	85	510	36	7225
118	$\Sigma = 805$	$\Sigma = 9414$	$\Sigma = 1540$	$\Sigma = 64861$

$$n = 10$$

Formula:-

$$r^2 = \frac{(10)(9414) - (118)(805)}{[10(1540) - (118)^2][10(64861) - (805)^2]}$$

$$[10(1540) - (118)^2][10(64861) - (805)^2]$$

$$= \frac{94140 - 94990}{(18400 - 13924)(648610 - 648025)}$$

$$= \frac{-850}{(1476)(585)} \Rightarrow \frac{-880}{863460}$$

$$\gamma^2 = -0.0009$$

Problem No 4:-

X	Y	XY	X ²	Y ²
8	8	64	64	64
4	3	12	16	9
2	4	8	4	16
3	7	21	9	49
5	2	10	25	4
5	0	0	25	0
$\Sigma X = 18$	$\Sigma Y = 30$	$\Sigma XY = 73$	$\Sigma X^2 = 64$	$\Sigma Y^2 = 202$

apply formula:-

$$r^2 = \frac{n \Sigma XY - (\Sigma X)(\Sigma Y)}{[n \Sigma X^2 - (\Sigma X)^2][n \Sigma Y^2 - (\Sigma Y)^2]}$$

$$= \frac{6(73) - (18)(30)}{[6(64) - 324][6(202) - 900]}$$

$$= \frac{426 - 540}{(384 - 324)(1212 - 900)}$$

$$= \frac{-114}{(60)(312)} \rightarrow \frac{-114}{18720}$$

$$r^2 = -0.006$$

Problem No 5:-

formula:- $r^2 = \frac{n \Sigma XY - (\Sigma X)(\Sigma Y)}{[n \Sigma X^2 - (\Sigma X)^2][n \Sigma Y^2 - (\Sigma Y)^2]}$

X	Y	XY	X ²	Y ²
1	89	89	1	7921
12	76	912	144	5776
6	52	312	36	2704
10	56	560	100	3136
9	51	459	81	2601
10	11	110	100	121
1	57	57	1	3249
8	55	440	64	3025
12	67	804	144	4489
6	53	318	36	2809
11	78	858	121	6084
8	89	712	64	7921

$$\Sigma Y = 156 \quad \Sigma X^2 = 103 \quad \Sigma XY = 4960$$

$$\Sigma X^2 = 1030 \quad \Sigma Y^2 = 43520 \quad n = 12$$

substituting values:-

$$= \frac{12(4960) - (103)(156)}{[12(1030) - (103)^2][12(43520) - (156)^2]}$$

$$= \frac{59520 - 16068}{[12240 - 10609][52224 - 24336]}$$

$$= \frac{1318}{(576)(1188)}$$

$$= \frac{1318}{683648} \rightarrow r^2 = 0.002 \quad \text{Ans.}$$

to find $r =$
taking square root on b1s.

$$r^2 = 0.0002$$

$$\sqrt{r^2} = \sqrt{0.0002}$$

$$r = 0.014$$

Answer